Systems of inequalities. Number intervals

Two conditions at once — the answer is where the two solution sets overlap.

LESSON 51–52 | 2 × 40 MIN | ALGEBRA 8, §16 | STAGE 9 · 4.3

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Solve a system of two linear inequalities.
- Find the intersection of two intervals on the number line.
- Use the standard names and notation for intervals.
- Solve a double inequality such as $-2 < 3x + 1 \le 10$.

TEXTBOOK

National	§16, pp. 94–104
Cambridge	Learner’s Book pp. 96–102
Workbook	Workbook 4.3

01 Explanation

The names of the intervals

CONDITION	NOTATION	NAME
$a \le x \le b$	$[a; b]$	segment (closed interval)
$a < x < b$	$(a; b)$	open interval
$a \le x < b$	$[a; b)$	half-open interval
$x \ge a$	$[a; +\infty)$	closed ray
$x < a$	$(-\infty; a)$	open ray

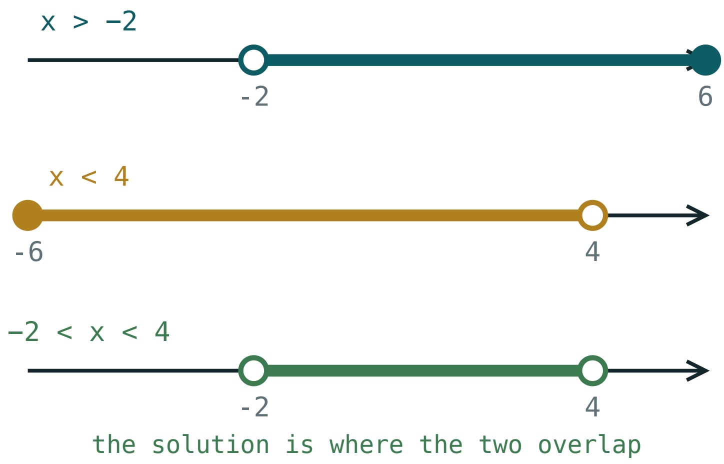
Square bracket = the end belongs to the set (filled circle). Round bracket = it does not (open circle). Infinity is always round.

Solving a system

METHOD

1. Solve each inequality on its own.

2. Draw both solution sets on the **same** number line.
3. The answer is the part shaded **twice** — the intersection.



Each inequality shades part of the line; the system is satisfied only where both shadings overlap.

The word to say in class is **“and”**. A system asks for the values satisfying the first inequality *and* the second, so only the overlap counts.

SOMETIMES THERE IS NO OVERLAP

The system $x > 5$ and $x < 2$ has **no solutions** — the empty set. Say so; do not write a nonsense interval like $(5; 2)$.

Double inequalities

$-2 < 3x + 1 \le 10$ is a system written compactly. Work on all three parts at once:

$$-2 < 3x + 1 \le 10$$

$$-3 < 3x \le 9 \ \ \text{(subtract 1 throughout)}$$

$$-1 < x \le 3 \ \ \text{(divide by 3 throughout)}$$

The answer is the half-open interval $(-1; 3]$.

IF YOU DIVIDE BY A NEGATIVE

Both signs reverse and the two ends swap places, so rewrite the smaller number on the left.

02 Worked examples

EXAMPLE 1 ***Solve the system: $2x - 1 > 3$ and $x + 4 < 9$***

1. $2x > 4$, so $x > 2$

   The first inequality.

2. $x < 5$

   The second.

3. Draw both: shaded right of 2, and left of 5.

4. $2 < x < 5$

   The overlap.

ANSWER

$(2; 5)$

EXAMPLE 2 ***Solve $-2 < 3x + 1 \le 10$***

1. $-3 < 3x \le 9$

   Subtract 1 from all three parts.

2. $-1 < x \le 3$

   Divide all three by 3.

3. $(-1; 3]$

   Open at −1, closed at 3.

ANSWER

$(-1; 3]$

EXAMPLE 3 ***Solve the system: $x \geq 4$ and $2x < 6$***

1. $x \geq 4$

   The first.

2. $x < 3$

   The second.

3. Nothing is both at least 4 and less than 3.

4. No solutions — the empty set.

ANSWER

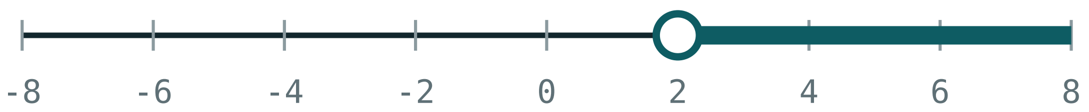
No solutions.

03 Interactive model

Set each boundary in turn and ask the class which part of the line survives both.

One inequality at a time

$$x > 2$$



inequality `x > 2` | boundary `2 (open)` | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
System of inequalities	Tengsizliklar sistemasi	Система неравенств
Intersection	Kesishma	Пересечение
Number interval	Sonli oraliq	Числовой промежуток
Segment (closed interval)	Kesma	Отрезок
Open interval	Ochiq oraliq	Интервал
Half-open interval	Yarim ochiq oraliq	Полуинтервал
Ray (half-line)	Nur	Луч
Double inequality	Qo‘sh tengsizlik	Двойное неравенство
Empty set	Bo‘sh to‘plam	Пустое множество

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$[2; 7)$ means:

- $2 < x < 7$
- $2 \le x < 7$
- $2 < x \le 7$
- $2 \le x \le 7$

The system $x > 1$ and $x < 6$ gives:

- $(1; 6)$
- $(-\infty; 6)$
- $(1; +\infty)$
- no solutions

The system $x > 5$ and $x < 2$ gives:

- $(2; 5)$
- $(5; 2)$
- no solutions
- every number

$-4 \le 2x < 6$ gives:

- $[-2; 3)$
- $(-2; 3]$
- $[-4; 6)$
- $[-2; 3]$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Write $3 \le x \le 8$ as an interval*

   ANSWER $[3; 8]$

2. *Write $0 < x < 5$ as an interval*

   ANSWER $(0; 5)$

3. *Write $x \ge 2$ as an interval*

   ANSWER $[2; +\infty)$

4. *What does $(-1; 4]$ mean?*

   ANSWER $-1 < x \le 4$

5. *Solve the system $x > 1$ and $x < 6$*

   ANSWER $(1; 6)$

6. *Solve the system $x \ge 0$ and $x \le 3$*

   ANSWER $[0; 3]$

7. *Solve the system $x > 4$ and $x > 7$*

   ANSWER $(7; +\infty)$

Medium · the standard question

7 PROBLEMS

1. *Solve:* $2x - 1 > 3$ *and* $x + 4 < 9$

   ANSWER $(2; 5)$

2. *Solve:* $3x \le 12$ *and* $x + 1 > 0$

   ANSWER $(-1; 4]$

3. *Solve* $-2 < 3x + 1 \le 10$

   ANSWER $(-1; 3]$

4. *Solve* $-4 \le 2x < 6$

   ANSWER $[-2; 3)$

5. *Solve:* $x \ge 4$ *and* $2x < 6$

   ANSWER No solutions.

6. *Solve* $1 \le x - 3 \le 5$

   ANSWER $[4; 8]$

7. *Solve:* $5 - x > 1$ *and* $x + 2 \ge 0$

   ANSWER $[-2; 4)$

Hard · stretch and reason

7 PROBLEMS

1. *Solve* $-5 < 2 - 3x \le 8$

   ANSWER $-7 < -3x \le 6 \Longrightarrow -2 \le x < \frac{7}{3}$

2. *Solve:* $2(x - 1) \ge x$ *and* $3x < x + 8$

   ANSWER $x \ge 2$ and $x < 4$: $[2; 4)$

3. *How many whole numbers satisfy* $-3 < 2x + 1 \le 9$?

   ANSWER $-2 < x \le 4$: the numbers $-1, 0, 1, 2, 3, 4$ — six of them.

4. *Solve:* $\frac{x}{2} > 1$ *and* $\frac{x}{3} < 2$

   ANSWER $x > 2$ and $x < 6$: $(2; 6)$

5. *For which* a *does the system* $x > 3$ *and* $x < a$ *have solutions?*

   ANSWER $a > 3$

6. *Solve* $0 \le \frac{2x - 1}{3} \le 3$

   ANSWER $0 \le 2x - 1 \le 9 \Longrightarrow [0.5; 5]$

7. *Find all whole numbers satisfying both* $4x > 6$ *and* $3x \le 15$

   ANSWER $x > 1.5$ and $x \le 5$: $2, 3, 4, 5$

07 Homework

Homework — 6 problems

Algebra 8, §16, pp. 94–104. Draw both sets on one line before writing the answer.

1. *Solve:* $3x - 2 > 7$ *and* $x - 1 < 6$
2. *Solve* $-6 \le 4x + 2 < 10$
3. *Solve:* $x + 5 \ge 2$ *and* $2x \le 8$
4. *Solve:* $x > 6$ *and* $x < 3$*, and explain the answer*

5. *How many whole numbers satisfy* $-4 < 3x - 1 \le 11$*?*

6. *Write* $[-3; 5)$ *in words and draw it on a number line*